



LOB 1003 - Cálculo 1

Gabarito da Lista de exercícios 4

1. $y = -\frac{1}{4}x + 1.$

2. $y = -2x + 3.$

3. $y = \frac{1}{3}(x + 2)$

4. $y = 2x$

5.

a) $f'(x) = 6x$

b) $f'(x) = 9x^2 - 4x$

c) $f'(x) = -\frac{6}{x^3}$

d) $f'(x) = 3 - \frac{1}{x^2}$

e) $f'(x) = 2x^2 + \frac{x}{2}$

f) $f'(x) = 2 - \frac{1}{x^2} - \frac{2}{x^3}$

g) $f'(x) = 3x^2 + 2x$

h) $f'(x) = 3 + \frac{1}{\sqrt{x}}$

i) $f'(x) = \frac{2}{3\sqrt[3]{x^2}}$

j) $f'(x) = -\frac{4}{x^2} - \frac{10}{x^3}$

k) $f'(x) = \frac{1}{3\sqrt[3]{x^2}} + \frac{1}{2\sqrt{x}}$

l) $f(x) = 6x^3 + \sqrt[3]{x}$

6.

a) $f'(x) = 6x - 5 \sin x$

b) $f'(x) = \sin x + x \cos x$

c) $f'(x) = \frac{\operatorname{tg} x - (x + 1) \sec^2 x}{\operatorname{tg}^2 x}$

d) $f'(x) = \frac{\sec x [(3x + 2) \operatorname{tg} x - 3]}{(3x + 2)^2}$

e) $f'(x) = \sec x \left(\frac{1}{2\sqrt{x}} + \sqrt{x} \operatorname{tg} x \right)$

f) $f'(x) = \cotg x - x \operatorname{cosec}^2 x$

g) $f'(x) = 2x + 3(\operatorname{tg} x + x \sec^2 x)$

h) $f'(x) = \frac{-\sin x - x(x + 1) \cos x}{x^2 \sin^2 x}$

i) $f'(x) = \operatorname{cosec} x \left(3x^2 + \frac{1}{2\sqrt{x}} - (x^3 + \sqrt{x}) \cotg x \right)$

$$j) f'(x) = \frac{-(x^2 + 1) \operatorname{sen} x - 2x \cos x}{(x^2 + 1)^2}$$

$$k) f'(x) = 2x \operatorname{tg} x + x^2 \sec^2 x$$

$$l) f'(x) = \frac{-3(\cos x - \operatorname{sen} x)}{(\operatorname{sen} x + \cos x)^2}$$

$$m) f'(x) = (2x - 1) \operatorname{sen} x + (x^2 + 1) \cos x$$

$$n) f'(x) = -3 \operatorname{sen} x + 5 \sec x \operatorname{tg} x$$

$$o) f'(x) = 4 \sec x \operatorname{tg} x - \operatorname{cosec}^2 x$$

$$p) f'(x) = \frac{2x - (x^2 + 1) \operatorname{tg} x}{\sec x}$$

$$q) f'(x) = \frac{1 + x \cot x}{\operatorname{cosec} x}$$

$$r) f'(x) = \frac{(x - 1) \cos x - (x + 1) \operatorname{sen} x - 1}{(x - \cos x)^2}$$

7.

$$a) y' = x e^x (2 + x)$$

$$b) y' = e^x (\cos x - \operatorname{sen} x)$$

$$c) y' = x(2 \ln x + 1) + 2e^x$$

$$d) y' = \frac{-x - \ln x - 1}{(x \ln x)^2}$$

$$e) y' = \frac{e^x (x - 2)}{x^3}$$

$$f) y' = \frac{1 - \ln x}{x^2}$$

$$g) y' = 3 + \frac{5}{x}$$

$$h) y' = \frac{2e^x}{(1 - e^x)^2}$$

$$i) y' = x^3 e^x (4 + x)$$

$$j) y' = e^x (\operatorname{sen} x \cos x + \cos^2 x - \operatorname{sen}^2 x)$$

$$k) y' = 5x(2 \ln x + 1)$$

$$l) y' = \frac{e^x (x - 1)^2}{(x^2 + 1)^2}$$

$$m) y' = \frac{x e^x}{(x + 1)^2}$$

$$\text{n) } y' = e^x \left(\frac{2x+1}{2\sqrt{x}} \right)$$

$$\text{o) } y' = \frac{-ACe^x}{(B+Ce^x)^2}$$

8.

$$\text{a) } y' = -2x, y'' = -2$$

$$\text{b) } s' = 15t^2 - 15t^4, s'' = 30t - 60t^3$$

$$\text{c) } y' = 4x^2 - 1 + 2e^x, y'' = 8x + 2e^x$$

$$\text{d) } w' = -6z^{-3} + z^{-2}, w'' = 18z^{-4} - 2z^{-3}$$

$$\text{e) } p' = \frac{q}{6} + \frac{q^{-3}}{6} + q^{-5}, p'' = \frac{1}{6} - \frac{q^{-4}}{2} - 5q^{-6}$$

$$\text{f) } r' = -\frac{2}{3}s^{-3} + \frac{5}{2}s^{-2}, r'' = 2s^{-4} - 5s^{-3}$$

$$\text{g) } y' = 2x - 7x^{-2}, y'' = \frac{2x^3 + 14}{x^3}$$

$$\text{h) } r' = 3\theta^{-4}, r'' = -15\theta^{-5}$$

$$\text{i) } w' = 6ze^z + 3z^2e^z, w'' = 6e^z + 12ze^z + 3z^2e^z$$

$$\text{j) } y' = 12x - 10 + 10x^{-3}, y'' = 12 - 30x^{-4}$$

9.

$$\text{a) } y' = 10(2x+1)^4$$

$$\text{b) } y' = \left(1 - \frac{x}{7} \right)^{-8}$$

$$\text{c) } y' = 4 \left(\frac{x^2}{8} + x - \frac{1}{x} \right)^3 \left(\frac{x}{4} + 1 + \frac{1}{x^2} \right)$$

$$\text{d) } y' = \sec(\operatorname{tg} x) \operatorname{tg}(\operatorname{tg} x) \sec^2 x$$

$$\text{e) } y' = 3 \operatorname{sen}^2 x \cos x$$

$$\text{f) } y' = -5e^{-5x}$$

$$\text{g) } y' = -7e^{5-7x}$$

$$\text{h) } p' = -\frac{1}{2\sqrt{3-t}}$$

$$\text{i) } s' = \frac{4}{\pi}(\cos(3t) - \operatorname{sen}(5t))$$

$$\text{j) } r' = \frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta + \operatorname{cotg} \theta}$$

$$\text{k) } y' = 2x \operatorname{sen}^4 x + 4x^2 \operatorname{sen}^3 x \cos x + \cos^{-2} x + 2x \operatorname{sen} x \cos^{-3} x$$

$$\text{l) } y' = (1-x)e^{-x} + 3e^{3x}$$

$$\text{m) } y' = e^{5x/2} \left(\frac{5}{2}x^2 - 3x + 3 \right)$$

$$\text{n) } f'(\theta) = \frac{2 \operatorname{sen} \theta}{(1 + \cos \theta)^2}$$

$$\text{o) } r' = 2(\theta \cos \theta^2 \cos(2\theta) - \operatorname{sen} \theta^2 \operatorname{sen}(2\theta))$$

$$\text{p) } y' = 2\theta e^{-\theta^2} \operatorname{sen}(e^{-\theta^2})$$

$$\text{q) } y' = -2 \operatorname{sen}(2t-5) \cos(\cos(2t-5))$$

$$\text{r) } y' = \frac{-t \operatorname{sen} t^2}{\sqrt{1 + \cos t^2}}$$

10.

$$\text{a) } F'(x) = 7(x^3 + 4x)^6(3x^2 + 4)$$

$$\text{b) } y' = -3x^2 \operatorname{sen}(a^3 + x^3)$$

$$\text{c) } h'(t) = 4t^2(t^3 + 1)^3(4t^4 + t - 3)$$

$$\text{d) } y' = e^{x \cos x}(\cos x - x \operatorname{sen} x)$$

$$\text{e) } y' = -\operatorname{sen} x \sec^2(\cos x)$$

$$\text{f) } y' = -2 \cos \theta \cot g(\operatorname{sen} \theta) \operatorname{cosec}^2(\operatorname{sen} \theta)$$

$$\text{g) } y = \frac{4e^{2x}}{(1 + e^{2x})^2} \operatorname{sen} \left(\frac{1 - e^{2x}}{1 + e^{2x}} \right)$$

$$\text{h) } F'(x) = \frac{2 + 3x^2}{4(1 + 2x + x^3)^{3/4}}$$

$$\text{i) } y' = (1 - kx)e^{-kx}$$

$$\text{j) } y' = 8(2x-5)^3(8x^2-5) - 48x(2x-5)^4(8x^2-5)^{-4}$$

$$\text{k) } F'(z) = \frac{1}{(z-1)^{1/2}(z+1)^{3/2}}$$

$$\text{l) } y' = 4 \operatorname{tg} x \sec^2 x$$

$$\text{m) } f'(t) = e^t \sec^2(e^t) + e^{\operatorname{tg} t} \sec^2 t$$

$$\text{n) } y' = -\frac{\pi \operatorname{sen}(\sqrt{\operatorname{sen}(\operatorname{tg}(\pi x))}) \cos(\operatorname{tg}(\pi x)) \sec^2(\pi x)}{2 \sqrt{\operatorname{sen}(\operatorname{tg}(\pi x))}}$$

$$\text{o) } g'(t) = \frac{-12t^3}{(t^4 + 1)^4}$$

$$\text{p) } g'(x) = 20(1+4x)^4(3+x-x^2)^8 + 8(1-2x)(1+4x)^5(3+x-x^2)^7$$

$$\text{q)} \quad y' = \frac{-12x(x^2 + 1)^2}{(x^2 - 1)^4}$$

$$\text{r)} \quad y' = \frac{1}{(r^2 + 1)^{3/2}}$$

$$\text{s)} \quad g'(x) = 2r^2 p e^{rx} (2r e^{rx} + n)^{p-1}$$

$$\text{t)} \quad f'(t) = 4e^{\sin^2 t} \sin t \cos t \sin(e^{\sin^2 t}) \cos(e^{\sin^2 t})$$

$$\text{u)} \quad y' = e^{\alpha x} (\alpha \sin(\beta x) + \beta \cos(\beta x))$$

11.

$$\text{a)} \quad y = -\frac{3}{16}x + \frac{11}{4}$$

$$\text{b)} \quad y = -x + \pi$$